

SAFWEN BEN MABROUK

Full-Stack Software Engineer | Next.js • NestJS • TypeScript • React

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PROFILE

Full-stack software engineer specializing in Next.js, NestJS, and TypeScript. I focus on building end-to-end web applications that solve real problems from designing multi-workspace data isolation and real-time collaboration, to integrating machine learning routing pipelines. I pride myself on writing clean, secure business logic and shipping reliable software through automated testing and Docker deployment.

EDUCATION

Software Engineering Degree 2023 – 2026

EPI – International Multidisciplinary School, Sousse

Preparatory Cycle in Technology and Computer Science (TIC) 2021 – 2023

EPI – International Multidisciplinary School, Sousse

Baccalaureate in Technical Sciences 2019

Bourguiba High School, Monastir

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern — Trinovatech February 2026 – May 2026

Monastir

- Led the end-to-end development of a full-stack web platform during a software engineering internship, following Agile methodologies.
- Structured the database models and core business logic to support complex data relationships and user workflows.
- Increased software reliability through automated unit and integration testing, Docker containerization, and CI/CD pipelines.

Full-Stack Developer Intern — SWConsulting June 2025 – July 2025

Monastir

- Analyzed a microservice-based production system (Next.js, NestJS, PostgreSQL) used by banking clients to identify business logic bottlenecks.
- Decoupled financial estimation rules from the core source code, enabling non-technical teams to modify calculation logic dynamically without requiring code changes or redeployments.
- Delivered full-stack features in a collaborative Agile environment.

PROJECTS

DocuMind — Multi-Workspace Document Management Platform

Next.js • NestJS • PostgreSQL/pgvector • Prisma • RAG • Socket.io • Stripe • Docker

- Developed a multi-workspace document management platform featuring granular Role-Based Access Control (RBAC), isolating data and permissions per workspace, folder, and member.
- Integrated a RAG pipeline using pgvector for semantic search, enabling instant information retrieval across complex enterprise documents.
- Implemented real-time collaborative editing via Socket.io, managing concurrent state conflicts, and containerized the architecture with Docker.

AI-Powered Used Car Price Analyzer

Next.js • NestJS • FastAPI • Python • XGBoost • KNN

- Scraped and cleaned a dataset of 6,000+ Tunisian car listings to train custom machine learning models for price estimation and deal detection.
- Designed an intelligent routing system using KNN to dynamically select between two specialized XGBoost models (budget vs. mid-budget vehicles) based on input features.
- Integrated the ML pipeline into the web app via a FastAPI microservice, with a Next.js admin dashboard for monitoring predictions and usage.

Secure Medical Teleconsultation Platform

React • Node.js • Express • MongoDB • WebRTC • Stripe

- Developed a telemedicine platform using MVC architecture, supporting real-time WebRTC video consultations between doctors and patients.
- Built a strict onboarding flow where administrators must verify and approve doctors before they can access the platform and receive appointments.
- Implemented the end-to-end appointment lifecycle (booking, Stripe payment, consultation, and automated post-consultation medical record generation) alongside an admin dashboard for stats and user management.

Transactional E-Learning Platform (LMS)

Next.js • Node.js • Tailwind CSS • PostgreSQL • Prisma • Stripe

- Built a transactional LMS with strict role-based access (Teacher/Student), ensuring students can only view purchased course content.
- Implemented Stripe integration to handle course purchases, automatically unlocking content and tracking student learning progress (percentage completion) in real-time.
- Designed the database architecture to securely manage course materials, user purchases, and protected content delivery.

TECHNICAL SKILLS

Frontend: React.js, Next.js, Tailwind CSS, HTML5/CSS3
Backend: Node.js, Express.js, NestJS, REST APIs
Programming Languages: JavaScript, TypeScript, Python, Java, C, SQL
AI & Machine Learning: RAG, XGBoost, KNN, Data Scraping & Preprocessing
Databases & ORM: PostgreSQL, MongoDB, MySQL, Oracle, Prisma ORM, Firebase
Cloud & DevOps: AWS (Cloud Foundations), Docker, GitHub Actions (CI/CD), Vercel, Render
Tools & Methodologies: Git, Linux, Postman, Agile/Scrum

CERTIFICATIONS

- AWS Academy Graduate: Cloud Foundations, Dec 2024
- Cisco CCNA: Switching, Routing & Wireless Essentials, Jun 2024

LANGUAGES

Arabic (native) • English (advanced) • French (advanced)